NEET TEST HUB

NTH-PREMIUM TEST SERIES 04

Date – 05 October 2025	Duration – 3 Hours	Marks – 720

Physics :- System of Particles & Rotational Motion

Chemistry :- Structure of Atom

Biology :- Anatomy of Flowering Plants, Respiration in Plants, Locomotion & Movement, Neural Control & Coordination, Chemical Coordination & Integration

Instruction

1. The test is of 3 hours duration.

2. The test booklet consists of 200 questions. The maximum mark is 720.

3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,

4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).

5. There is only one correct response for each question.

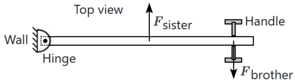
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.

7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.

8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

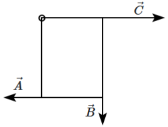
Physics

1. A brother and sister fight over whether a door should be opened or closed. They both push with the same magnitude of force, perpendicular to the door, as shown in the figure. Which sibling exerts more torque on the door, with respect to the hinge?



1.	The sister.
2.	The brother.
3.	They both exert the same torque.
4.	It depends upon the length of the door.

2. A rectangular piece of metal (0.3 m by 0.4 m) is hinged (⊗) as shown in the upper left corner, hanging so that the long edge is vertical. Force $\vec{A}$ (20 N) acts to the left at the lower left corner. Force $\vec{B}$ (10 N) acts down at the lower right corner. Force $\vec{C}$ (30 N) acts to the right at the upper right corner. (take counterclockwise to be positive)

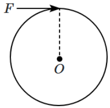


What is the torque of force $\vec{C}$ about the pivot?
1. -9 N-m
2. 0
3. 4.5 N-m
4. 9 N-m

3. A massless metre stick sits on a fulcrum at its 0.4 m mark. A 6 kg mass sits on the metre stick at 0.2 m mark. What mass is required to sit at the 0.9 m mark in order to have torque balance? (take $g = 10$ m/s^2)

1.	2.4 kg	2.	4.5 kg
3.	10 kg	4.	15 kg

4. A uniform disc of mass M and radius R is fixed, so that it is free to rotate in its own plane, about the centre O. A force F is applied tangentially to the disc, continuously, for one complete revolution, starting from rest.



The angular acceleration (α) of the disc is:

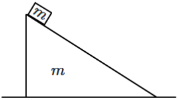
1.	$\dfrac{F}{2MR}$	2.	$\dfrac{F}{MR}$
3.	$\dfrac{2F}{MR}$	4.	$\dfrac{3F}{2MR}$

5. Given below are two statements:

Assertion (A):	For a system of particles under a central force field, the total angular momentum is conserved.
Reason (R):	The torque acting on such a system is zero.

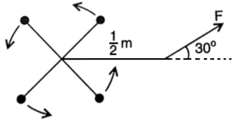
1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

6. A block of mass m is released on the inclined face of smooth wedge of equal mass m, which lies on a smooth horizontal plane. The system is initially at rest. When the block reaches the bottom of the wedge, and just before it hits the plane, it has a horizontal velocity component (w.r.t., ground) of v. The velocity of the wedge is:



1.	v	2.	$2v$
3.	$\frac{v}{2}$	4.	zero

7. Four 2 kg masses are connected by $\frac{1}{4}$ m spokes to an axle as in figure given below. A force F of 24 N acts on a lever $\frac{1}{2}$ m long to produce an angular acceleration α. The magnitude of α (in rad/s^2) is:



1.	2	2.	12
3.	6	4.	3

8. A particle is moving along a straight line with increasing speed. What happens to its angular momentum about a fixed point on this line?

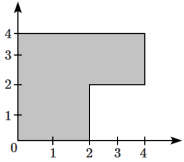
1.	It continuously increases.
2.	It continuously decreases.
3.	It may either increase or decrease depending on the direction of motion.
4.	It remains zero.

9. Given below are two statements:

Assertion (A):	The acceleration of the centre-of-mass of a system of particles does not depend on the internal forces between the particles.
Reason (R):	The net force acting on the system does not get any contribution from internal forces as the latter cancel in pairs, due to Newton's third law.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

10. A (2×2) square sheet is cut out from a (4×4) square sheet as shown in the figure. The shift in centre-of-mass would be:
(Assume length and breadth are measured in meters.)



1.	$\frac{\sqrt{2}}{3}$ m	2.	$2\sqrt{2}$ m
3.	$\frac{\sqrt{74}}{3}$ m	4.	1 m

11. The centre-of-mass of a thin, uniform triangular lamina lies at its:
1. orthocenter
2. circumcenter
3. centroid
4. incenter

12. The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at point (2, 0, −3) about the point (2, −2, −2) is given by:
1. $-8\hat{i} - 4\hat{j} - 7\hat{k}$
2. $-4\hat{i} - \hat{j} - 8\hat{k}$
3. $-7\hat{i} - 8\hat{j} - 4\hat{k}$
4. $-7\hat{i} - 4\hat{j} - 8\hat{k}$

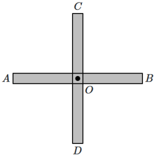
13. A string is wrapped along the rim of a wheel of the moment of inertia 0.10 kg-m^2 and radius 10 cm. If the string is now pulled by a force of 10 N, then the wheel starts to rotate about its axis from rest. The angular velocity of the wheel after 2 s will be:

1.	40 rad/s	2.	80 rad/s
3.	10 rad/s	4.	20 rad/s

14. The moment of inertia of a thin spherical shell of mass M and radius R about the diameter is:

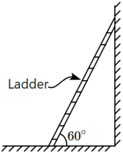
1.	$\frac{1}{2}MR^2$	2.	$\frac{1}{3}MR^2$
3.	$\frac{2}{5}MR^2$	4.	$\frac{2}{3}MR^2$

15. Two identical uniform rods of mass m, length L (each) are joined at their centres at right angles, as shown in the figure. The moment of inertia of the system about an axis passing through A, perpendicular to the plane of the rods is:



1.	$\frac{2}{3}mL^2$	2.	$\frac{5}{12}mL^2$
3.	$\frac{5}{6}mL^2$	4.	$\frac{7}{6}mL^2$

16. A ladder leaning against a wall makes an angle of 60° with the horizontal, as shown below. A person needs to climb this ladder.



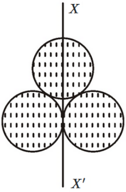
The ladder is more likely to slip when the person:

1.	just steps on the very bottom end of the ladder.
2.	stands on the ladder near the bottom end.
3.	stands on the ladder near the middle.
4.	stands on the ladder near the top end.

17. The ratio of the radius of gyration of a circular disc to that of a circular ring, both having the same mass and radius, about their respective axes is:

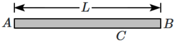
1.	$\sqrt{2} : \sqrt{3}$	2.	$\sqrt{3} : \sqrt{2}$
3.	$1 : \sqrt{2}$	4.	$\sqrt{2} : 1$

18. Three identical spherical shells, each of mass m and radius r are placed as shown in the figure. Consider an axis XX', which is touching two shells and passing through the diameter of the third shell. The moment of inertia of the system consisting of these three spherical shells about the XX' axis is:



1.	$\frac{11}{5}mr^2$	2.	$3mr^2$
3.	$\frac{16}{5}mr^2$	4.	$4mr^2$

19. Consider a linear mass distribution along a rod with the property that the linear mass density is proportional to the distance from one end A. At what distance from A will the centre-of-mass lie?



1.	$\frac{L}{2}$	2.	$\frac{L}{3}$
3.	$\frac{2L}{3}$	4.	$\frac{3L}{4}$

20. Given below are two statements:

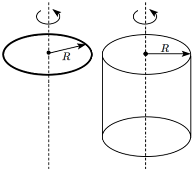
Statement I:	The motion of the centre-of-mass of a system of particles is not affected by internal forces within the system.
Statement II:	The internal forces within a system of particles constitute action-reaction pairs, and their vector sum is zero.

1.	**Statement I** is incorrect and **Statement II** is correct.
2.	Both **Statement I** and **Statement II** are correct.
3.	Both **Statement I** and **Statement II** are incorrect.
4.	**Statement I** is correct and **Statement II** is incorrect.

21. The moment of inertia of a thin uniform spherical shell and of a uniform solid sphere of the same masses and radii, taken about their respective diameters, are I_O (shell) and I_S (solid). Then:

1. $I_O = I_S$
2. $\frac{I_O}{2} = \frac{I_S}{3}$
3. $\frac{I_O}{3} = \frac{I_S}{5}$
4. $\frac{I_O}{5} = \frac{I_S}{3}$

22. What is the ratio of the radius of gyration of a solid sphere to that of a solid cylinder, given that both objects have the same mass and radius?



1.	$\frac{2}{\sqrt{5}}$	2.	$\frac{\sqrt{5}}{2}$
3.	$\sqrt{\frac{2}{5}}$	4.	$\frac{2}{\sqrt{3}}$

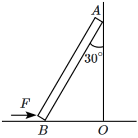
23. Two discs of the same moment of inertia are rotating about their regular axis passing through centre and perpendicular to the plane of the disc with angular velocities ω_1 and ω_2. They are brought into contact face to face with their axis of rotation coinciding. The expression for loss of energy during this process is:

1. $\frac{1}{4}I(\omega_1 - \omega_2)^2$
2. $I(\omega_1 - \omega_2)^2$
3. $\frac{1}{8}I(\omega_1 - \omega_2)^2$
4. $\frac{1}{2}I(\omega_1 - \omega_2)^2$

24. A disc rotating about its axis from rest acquires the angular speed 100 rev/s in 4 s. The angle rotated by it during these four seconds (in radians) is:

1.	100π	2.	200π
3.	300π	4.	400π

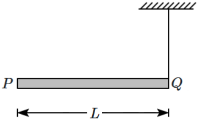
25. A uniform ladder of mass 10 kg is placed at an angle against a frictionless vertical wall, as shown in the figure, by applying a horizontal force F at the bottom (B) of the ladder, towards the wall. (Take $g = 10\ \text{m/s}^2$).



Assume that the ground is frictionless. The force F equals:

1.	$100\sqrt{3}$ N	2.	$50\sqrt{3}$ N
3.	$\dfrac{100}{\sqrt{3}}$ N	4.	$\dfrac{50}{\sqrt{3}}$ N

26. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in the figure. When the string is cut, the initial angular acceleration of the rod is:

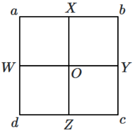


1.	$\dfrac{g}{L}$	2.	$\dfrac{2g}{L}$
3.	$\dfrac{2g}{3L}$	4.	$\dfrac{3g}{2L}$

27. The ratio of the radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of the same mass and radius about its axis is:
1. 5 : 2
2. 3 : 5
3. 5 : 3
4. 2 : 5

28. Two identical particles, each of mass m, are moving toward each other with velocities $2v$ and v. The velocity of their centre-of-mass will be:
1. v
2. $2v$
3. $\dfrac{v}{2}$
4. zero

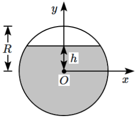
29. Consider a uniform square plate labelled *abcd* with a mass of 1 kg. Two point masses, each with a mass of 20 g, is placed at corners b and c as shown in the figure. Determine along which line the center-of-mass of the system shifts due to the placement of the point masses:

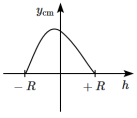


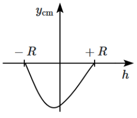
1.	OW	2.	OX
3.	OY	4.	OZ

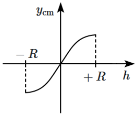
30. A motor that is switched on, accelerates uniformly from rest to a maximum speed of 600 rpm in 20 s. The angular acceleration is:
1. 30 rad/s^2
2. 30 rev/s^2
3. 1 rev/s^2
4. $\dfrac{1}{2}$ rev/s^2

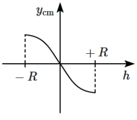
31. A thin uniform metallic spherical shell of mass $2m$ and radius R is half-filled with wax, and the amount of wax required has mass m. If it is completely filled, the amount of wax required is $2m$. The sphere is filled with wax until the level is at a height h above the centre O. The centre-of-mass of the entire system varies as a function of h. Which, of the following, correctly represents the y-coordinate (y_{cm}) of centre-of-mass as a function of h?



1.


2.


3.


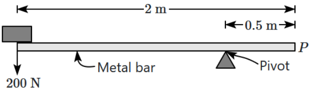
4.


32. Given below are two statements:

Assertion (A):	A ladder is more apt to slip when you are high up on it than when you just begin to climb.
Reason (R):	At high up on a ladder, the torque is large and on climbing up, the torque is small.

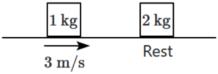
1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

33. A uniform bar with a length of 2.0 m and a weight of 1000 N has its centre of gravity at its centre (midpoint). The bar is pivoted as shown and supports a weight of 200 N at the position indicated in the diagram.



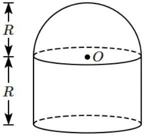
What weight is needed at position P to balance the bar?
1. 600 N
2. 800 N
3. 1000 N
4. 1600 N

34. A block of mass 1 kg moving with a velocity of 3 m/s undergoes a collision with a second block of mass 2 kg, at rest. After the collision, the two blocks move together.



The velocity of the centre-of-mass of the system before the collision is:
1. 3 m/s
2. 2 m/s
3. 1 m/s
4. zero

35. The centre-of-mass of a combination of a hemispherical shell and a cylindrical shell, both having the same height and radii and same mass, lies at a distance h from the centre of the hemisphere. Then, h equals:



1.	$\frac{R}{2}$	2.	$\frac{R}{\pi}$
3.	$\frac{2R}{\pi}$	4.	zero

36. Two identical projectiles are simultaneously launched into the air, with the same speed but in opposite directions – inclined at 60° to the horizontal. Ignore any air resistance. Their centre-of-mass:

1.	remains at rest
2.	moves along a horizontal straight line
3.	moves along a vertical straight line
4.	moves along a parabolic path

37. If two particles of masses 2 kg and 3 kg are placed at the two ends of a 1 m (light) rod, then the center-of-mass will be:

1.	40 cm from the 2 kg particle
2.	60 cm from the 3 kg particle
3.	60 cm from the 2 kg particle
4.	20 cm from the 3 kg particle

38. A rod of mass m length L rotates about one of its ends, in its own plane. The angular speed of the rod is ω. The angular momentum of the rod is:

1.	$\frac{1}{4}mL^2\omega$	2.	$\frac{1}{3}mL^2\omega$
3.	$\frac{2}{3}mL^2\omega$	4.	$\frac{1}{12}mL^2\omega$

39. A torque τ acts on a body of moment of inertia I rotating with angular speed ω. It will stop just after time:

1.	$\frac{I\tau}{\omega}$	2.	$\frac{I\omega}{\tau}$
3.	$\frac{\tau\omega}{I}$	4.	$I\omega\tau$

40. When a mass is rotating in a plane about a fixed point, its angular momentum is directed along:

1.	a line perpendicular to the plane of rotation
2.	the line making an angle of 45° to the plane of rotation
3.	the radius
4.	the tangent to the orbit

41. Given below are two statements:

Assertion (A):	If the same mass is drawn into the making of a solid sphere and a hollow sphere, the moment of inertia about the diameter of a hollow body is greater than that of a solid body.
Reason (R):	The moment of inertia depends on the mass distribution from the axis of rotation and in a hollow sphere, more of the mass is further away from the axis of rotation.

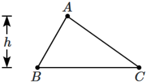
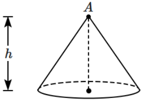
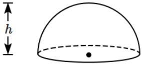
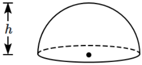
1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

42. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is:

1. 2×10^{6} N-m
2. 2×10^{-6} N-m
3. 2×10^{-3} N-m
4. 12×10^{-4} N-m

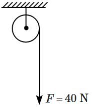
43. A force $\vec{F} = \alpha\hat{i} + 3\hat{j} + 6\hat{k}$ is acting at a point $\vec{r} = 2\hat{i} - 6\hat{j} - 12\hat{k}$. The value of α for which angular momentum about the origin is conserved is:
1. -1
2. 2
3. zero
4. 1

44. In **Column-I** are shown bodies of height (h) standing vertically upon their bases: in **Column-II** are given the height of the centre-of-mass of these bodies above their base, but in a different order. The heights are given as multiples of the height (h) in **Column-I**.

Column-I		Column-II	
(A)	 A uniform triangular lamina of height (altitude) h standing on its base BC.	(I)	$\frac{1}{2}$
(B)	 A uniform solid cone of height h.	(II)	$\frac{1}{3}$
(C)	 A thin uniform hemispherical bowl, standing on its open rim (as the base).	(III)	$\frac{1}{4}$
(D)	 A uniform solid hemisphere standing on its flat end (as the base).	(IV)	$\frac{3}{8}$

1.	A-II, B-II, C-III, D-IV
2.	A-II, B-I, C-III, D-IV
3.	A-III, B-II, C-I, D-III
4.	A-II, B-III, C-I, D-IV

45. The pulley has a radius of 50 cm and a moment of inertia of 0.5 kg-m^2. A frictional torque of 10 N-m acts on the axle. A constant force $F = 40$ N acts on the pulley through the light rope. The acceleration of the rope, as it unwinds, is (assume that the rope does not slip on the pulley)



1.	10 m/s^2	2.	5 m/s^2
3.	2.5 m/s^2	4.	20 m/s^2

Chemistry

46. Which of the following statements is/are not correct?

(a)	Isotones differ in the number of neutrons.
(b)	Isobars contain the same number of neutrons but differ in the number of protons.
(c)	No isotope of hydrogen is radioactive.
(d)	The density of the nucleus is much greater than that of an atom.

Choose the correct option:

1.	**(a)**, **(b)** and **(c)**	2.	**(b)**, **(c)** and **(d)**
3.	**(b)** and **(c)**	4.	All of the above

47. Which orbital has the maximum number of total nodes?
1. 5s
2. 5p
3. 5d
4. All have the same number of nodes.

48. The orbital having two radial as well as two angular nodes is:

1.	3p	2.	4f
3.	4d	4.	5d

49. A metal has a threshold frequency f_0. When light of frequency $v = 2f_0$ strikes the metal surface, the maximum velocity of the emitted electrons is v_1 When the frequency of the incident light is increased to $5f_0$, the maximum velocity of the emitted electrons becomes v_2. What is the ratio of v_1 to v_2 (i.e. $\frac{v_1}{v_2}$)?
1. 1 : 4
2. 1 : 2
3. 2 : 1
4. 4 : 1

50. Given below are two statement:

Statement I:	The energy of the He^+ ion in $n = 2$ state is same as the energy of H atom in $n = 1$ state.
Statement II:	It is possible to determine simultaneously the exact position and exact momentum of an electron in H atom.

1.	**Statement I** is incorrect and **Statement II** is correct.
2.	Both **Statement I** and **Statement II** are correct.
3.	Both **Statement I** and **Statement II** are incorrect.
4.	**Statement I** is correct and **Statement II** is incorrect.

51. A golf ball has a mass of 40g, and a speed of 45 m/s. If the speed can be measured within the accuracy of 2%, the uncertainty in the position is:
1. 14.6×10^{-23} m
2. 14.6×10^{-33} m
3. 1.46×10^{-23} m
4. 1.46×10^{-33} m

52. The energy of an electron in the first Bohr orbit of H atom is –13.6 eV. The possible energy value (s) of the excited state(s) for electrons in Bohr orbits of hydrogen is (are):
1. –3.4 eV
2. –4.2 eV
3. –6.8 eV
4. +6.8 eV

53. The potential energy of an electron present in Li^{+2} is:
1. $\frac{e^2}{2\pi\varepsilon_0 r}$
2. $\frac{3e^2}{4\pi\varepsilon_0 r}$
3. $\frac{-2e^2}{4\pi\varepsilon_0 r}$
4. $\frac{-3e^2}{4\pi\varepsilon_0 r}$

54. Consider the following statements:

Statement I:	The number of emitted photoelectrons increases with increase in the frequency of incident light.
Statement II:	The kinetic energy of emitted photoelectrons increases with increase in the frequency of incident light.

1.	**Statement I** is true but **Statement II** is false.
2.	**Statement I** is false but **Statement II** is true.
3.	Both **Statement I** and **Statement II** are true.
4.	Both **Statement I** and **Statement II** are false.

55. Which of the following orbits has a radius identical to that of the first Bohr orbit of a hydrogen atom?
1. $He^+ (n = 2)$
2. $Li^{2+} (n = 2)$
3. $Li^{2+} (n = 3)$
4. $Be^{3+} (n = 2)$

56. The lowest value of n that allows h orbitals to exist is:
1. 3
2. 4
3. 5
4. 6

57. If:
(i) energy of the subshell depends only on 'n' instead of 'n + l'
(ii) each orbital can accommodate a maximum of up to one electron
then for an element having an atomic number is 33, the correct statement is:
1. The group number is 5 and the period number is 4.
2. The period number is 5.
3. The last shell has only 1 electron.
4. The last electron is in the 4th shell.

58. Which of the following statements about cathode ray discharge tubes is correct?

1.	The electrical discharge can only be observed at high pressure and at low voltage.
2.	In the absence of an external electrical and magnetic field, cathode rays travel in straight lines.
3.	The characteristics of cathode rays depend upon the material of the electrodes.
4.	The characteristics of cathode rays depend upon the gas present in the cathode ray tube.

59. If a_0 is the radius of the H-atom and the de-Broglie wavelength of e^- in 3^{rd} orbit of Li^{2+} ion is $x\pi a_0$, then the value of x is:

1.	2	2.	4
3.	6	4.	8

60. An element with mass number 81 contains 31.7 % more neutrons as compared to protons. The atomic symbol of the element is:
1. ${}^{81}_{40}X$
2. ${}^{81}_{35}X$
3. ${}^{31.7}_{81}X$
4. ${}^{81}_{41}X$

61. Of the following transitions in hydrogen atom, the one which gives an absorption line of lowest frequency is:
1. n = 1 to n = 2
2. n = 3 to n = 8
3. n = 2 to n = 1
4. n = 8 to n = 3

62. 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is:

1.	5f > 6p > 4d > 5p	2.	5f > 6p > 5p > 4d
3.	6p > 5f > 5p > 4d	4.	6p > 5f > 4d > 5p

63. How many nodal planes are present in 4 dz^2 ?
1. 1
2. 2
3. 3
4. Zero

64. The orbital angular momentum value for 3s orbital is:

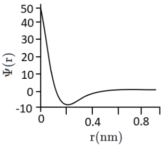
1.	0	2.	1
3.	2	4.	–1

65. The number of electrons that can be present in the subshells having m_s value of $-\frac{1}{2}$ for n = 4 are:

1.	36	2.	4
3.	16	4.	2

66. Given that the threshold frequency of the metal is $1.4 \times 10^{15} sec^{-1}$, what is the minimum energy required to eject a photoelectron from the metal?
[given: $h = 6.6 \times 10^{-34} Jsec$]
1. $9.24 \times 10^{-19} J$
2. $9.24 \times 10^{-18} J$
3. $4.62 \times 10^{-19} J$
4. $4.62 \times 10^{-18} J$

67. The graph represents the variation of the wave function with the distance from the nucleus.



The atomic orbital represented by above-mentioned graph is:

1.	1s	2.	2s
3.	2p	4.	None of the above

68. A microscope using appropriate photons is employed to locate an electron in an atom within a distance of 0.1 Å. Calculate the uncertainty involved in the measurement of its velocity.

1.	5.79×10^{6} m s^{-1}	2.	57.9×10^{6} m s^{-1}
3.	5.79×10^{7} m s^{-1}	4.	57.9×10^{7} m s^{-1}

69. Consider the given two statements:

Assertion (A):	Hydrogen atom has only one electron in its orbit but it produces several spectral lines.
Reason (R):	There are many available excited energy levels.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	Both **(A)** and **(R)** are False.

70. The most probable radius (in pm) for finding the electron in He^+ is :

1.	26.5	2.	105.8
3.	0.0	4.	52.9

71. With a ground state energy of –13.6 eV for a hydrogen atom, what would be the energy in the second excited state?
1. –6.8 eV
2. –3.4 eV
3. –1.51 eV
4. –4.53 eV

72. According to the law of photochemical equivalence, the energy absorbed (in ergs/mole) is given as:
($h = 6.62 \times 10^{-27}$ ergs, $c = 3 \times 10^{10}$ cm s^{-1}, $N_A = 6.02 \times 10^{23}$ mol^{-1})

1.	$\frac{1.196\times10^{8}}{\lambda}$	2.	$\frac{2.859\times10^{5}}{\lambda}$
3.	$\frac{2.859\times10^{16}}{\lambda}$	4.	$\frac{1.196\times10^{16}}{\lambda}$

73. What is the correct family and electronic configuration for element Z = 114?
1. Halogen family, $[Rn]5f^{14}6d^{10}7s^{2}7p^{5}$
2. Carbon family, $[Rn]5f^{14}6d^{10}7s^{2}7p^{2}$
3. Oxygen family, $[Rn]5f^{14}6d^{10}7s^{2}7p^{4}$
4. Nitrogen family, $[Rn]5f^{14}6d^{10}7s^{2}7p^{6}$

74. The number of unpaired electrons in Ni and Ni^{2+}, respectively, are (Z of Ni = 28)
1. 2, 2
2. 2, 4
3. 2, 0
4. 4, 2

75. The magnetic moment of Ni^{2+} (Z=28) is:
1. 1.732 BM
2. 2.82 BM
3. 3.87 BM
4. 4.89 BM

76. If the value of m for an electron is +3, it may be found in :

1.	4s orbital	2.	4p orbital
3.	In any f-orbital	4.	In any d-orbital

77. Given below are two statements:

Assertion (A):	In an atom, the velocity of electrons in the higher orbits keeps on decreasing.
Reason (R):	The velocity of the electron is inversely proportional to the radius of the orbit.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

78. The ratio of radii of 2nd & 3rd Bohr's orbit of H-atom is:
1. 2 : 3
2. 3 : 2
3. 4 : 9
4. 9 : 4

79. A pair of d-orbitals having electron density along the axes is:
1. d_{z^2}, d_{xz}
2. d_{xz}, d_{zy}
3. d_{z^2}, $d_{x^2-y^2}$
4. d_{xy}, $d_{x^2-y^2}$

80. What is the maximum number of electrons that can be associated with the following set of quantum numbers?
n = 3, l = 1 and m = –1.
1. 6
2. 4
3. 2
4. 10

81. The energy of electron in the ground state $(\mathrm{n}=1)$ for He^+ ion is -xJ, then that for an electron in $\mathrm{n}=2$ state for Be^{3+} ion in J is :

1.	$-\dfrac{x}{9}$	2.	$-4x$
3.	$-\dfrac{4}{9}x$	4.	$-x$

82. Incorrect statement among the following is:

1.	The uncertainty principle is$\Delta x \cdot \Delta p \geq \dfrac{h}{4\pi}$
2.	Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry, and a more balanced arrangement.
3.	The energy of the 2s orbital is less than the energy of the 2p orbital in the case of hydrogen-like atoms.
4.	De-Broglie's wavelength is given by$\lambda=\dfrac{h}{mv}$, where m= mass of the particle, v = group velocity of the particle.

83. Using the following statements, identify the correct set of statements:

(i).	n (principal quantum number) can have values 1, 2, 3, 4, ...
(ii).	The number of orbitals for a given value of l is (2l+1).
(iii).	The value of spin quantum numbers is always $\pm\frac{1}{2}$.
(iv).	For l=5, the total number of orbitals is 9.

1. (i), (ii), (iii)
2. (i), (ii), (iv)
3. (i), (ii), (iii), (iv)
4. (i), (iii), (iv)

84. Given the wavelengths of different radiations:
λ(A) = 300 nm , λ (B) = 300 μm , λ(C) = 3 nm , λ (D) = 30 A^o.
What is the correct sequence of these radiations in increasing order of their energies?

1.	B < C < A = D	2.	B < A < C = D
3.	A < D < C = B	4.	B < A < C < D

85. The azimuthal quantum number (l) is defined, as:

1.	Shape of orbitals	2.	Orientation of orbitals
3.	Energy of orbitals	4.	Size of orbitals

86. According to Bohr's model, how does the radius of the third orbit compare to the radius of the first orbit?
1. Equal to the radius of the first orbit
2. Three times the radius of the first orbit
3. Five times the radius of the first orbit
4. Nine times the radius of the first orbit

87. Read the following statements and choose the correct option.

Statement I:	If the ionisation potential of the hydrogen atom is 13.6 eV, the first excited state energy of He^+ ion will be 13.6 eV.
Statement II:	According to the prediction of simple molecular orbital theory, the bond orders of the He_2^+ ion and the H_2^+ ion are the same.

1. Both **statements** are correct.
2. Both **statements** are incorrect.
3. **Statement I** is correct and **Statement II** is incorrect.
4. **Statement II** is correct and **Statement I** is incorrect.

88. The relation between n_m (n_m = number of permissible values of magnetic quantum number (m_l) for a given value of azimuthal quantum number (l) is:
1. $n_m=l+2$
2. $l=\dfrac{n_m-1}{2}$
3. $l=2n_m+1$
4. $n_m=2l^2+1$

89. Which of the following electronic configurations for ground state atoms is incorrect?

1.	Co = [Ar] 4s: ↑↓ 3d: ↑↓ ↑↓ ↑ ↑ ↑
2.	Ni = [Ar] 4s: ↑↓ 3d: ↑↓ ↑↓ ↑↓ ↑ ↑
3.	Cu = [Ar] 4s: ↑↓ 3d: ↑↓ ↑↓ ↑↓ ↑↓ ↑
4.	Zn = [Ar] 4s: ↑↓ 3d: ↑↓ ↑↓ ↑↓ ↑↓ ↑↓

90. Given below are two statements:

Statement I:	Line emission spectra are useful in the study of electronic structure.
Statement II:	Each element has a unique line emission spectrum.

1.	**Statement I** is incorrect and **Statement II** is correct.
2.	Both **Statement I** and **Statement II** are correct.
3.	Both **Statement I** and **Statement II** are incorrect.
4.	**Statement I** is correct and **Statement II** is incorrect.

Biology

91. A steroid hormone when released increases blood glucose level, causes lipolysis and proteolysis, retards cellular uptake and utilization of amino acids by body cells and has a potent anti-inflammatory effect. This hormone is:
1. Thyroxin
2. Adrenaline
3. Growth hormone
4. Cortisol

92. Oxidation of acetyl-CoA via the TCA cycle requires:

I:	The continued replenishment of oxaloacetic acid.
II:	The regeneration of NAD^+ and FAD.

1. Only **I** is correct
2. Only **II** is correct
3. Neither **I** nor **II** is correct
4. Both **I** and **II** are correct

93. How many of the given statements are correct?

I:	In males, LH stimulates the synthesis and secretion of hormones called androgens from testis.
II:	In males, FSH and androgens regulate spermatogenesis.
III:	In females, LH induces ovulation of fully mature follicles (graafian follicles) and maintains the corpus luteum, formed from the remnants of the graafian follicles after ovulation.
IV:	FSH stimulates growth and development of the ovarian follicles in females.

1.	1	2.	2
3.	3	4.	4

94. Consider the given statements:

I:	The human skeletal system consists of 206 bones.
II:	The femur is the longest bone in the human body.
III:	Cartilaginous joints are immovable.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

95. Consider the given two statements:

I:	At maturity, none of the components of xylem tissue is living.
II:	At maturity, none of the components of phloem tissue is dead.

The correct statement/s is/are:
1. Only **I**
2. Only **II**
3. Both **I** and **II**
4. Neither **I** nor **II**

96. How do thyroid hormones exert their effects on target cells?

1.	By binding to membrane-bound receptors and activating second messenger pathways
2.	By binding to intracellular receptors and influencing gene expression
3.	By stimulating the release of neurotransmitters in the synaptic cleft
4.	By binding directly to mitochondrial receptors and altering ATP production

97. Which of these statements is incorrect?

1.	Enzymes of the TCA cycle are present in the mitochondrial matrix.
2.	Glycolysis occurs in the cytosol.
3.	Glycolysis operates as long as it is supplied with NAD that can pick up hydrogen atoms.
4.	Oxidative phosphorylation takes place in the outer mitochondrial membrane.

98. All the following regarding glycolysis are true except:

1.	It is often referred to as the EMP pathway.
2.	In anaerobic organisms, it is the only process in respiration.
3.	Glycolysis occurs in the cytoplasm of the cell and is present in all living organisms.
4.	In glycolysis, glucose undergoes reduction to form two molecules of pyruvic acid.

99. In the presence of oxygen, pyruvic acid is converted to Acetyl-CoA. During this metabolic conversion, what happens to pyruvic acid?
1. It gets oxidized.
2. It gets reduced.
3. It gets isomerized.
4. It is broken down into carbon dioxide.

100. The QRS complex in a standard ECG represents:
1. Depolarisation of auricles
2. Depolarisation of ventricles
3. Repolarisation of ventricles
4. Repolarisation of auricles

101. The epidermal tissue system in plants:

I:	forms the outer-most covering of the whole plant body.
II:	comprises epidermal cells, stomata and the epidermal appendages – the trichomes and hairs.
III:	is usually multi-layered.
IV:	has parenchymatous cells without a vacuole.

1. Only **I** and **II** are correct
2. Only **III** and **IV** are correct
3. Only **I** and **III** are correct
4. Only **II** and **IV** are correct

102. Enlargement of the thyroid gland, commonly called goitre is mainly caused by the deficiency of which of the following minerals in our diet:
1. Calcium
2. Phosphorus
3. Iodine
4. Potassium

103. In which tissue and under what conditions is sclerenchyma primarily found, and what function does it perform?

1.	Found in young stems and roots, providing flexibility and photosynthesis.
2.	Located near the epidermis, aiding in the regulation of transpiration and gaseous exchange.
3.	Present in seed coats, nut shells, and vascular bundles; composed of dead cells with thick lignified walls that provide mechanical support.
4.	Found in leaf mesophyll, enabling photosynthesis through chloroplasts.

104. A healthy male has a resting cardiac output of 6000 ml per minute with a stroke volume of 80 ml. What is the heart rate of this person?

1.	72 per minute	2.	75 per minute
3.	80 per minute	4.	100 per minute

105. Identify the correct statements:

I:	Arteries take blood away from the heart.
II:	Veins carry blood towards the heart.

1. Only **I**
2. Only **II**
3. Both **I** and **II**
4. Neither **I** nor **II**

106. Identify the correct statements:

I:	Binding of a hormone to its receptor leads to the formation of a hormone-receptor complex.
II:	Each receptor is specific to one hormone only and hence receptors are specific.
III:	Hormone-Receptor complex formation leads to certain biochemical changes in the target tissue.

1. Only **I** and **II**
2. Only **I** and **III**
3. Only **II** and **III**
4. **I**, **II** and **III**

107. Match the following blood components with their functions:

	Component (List I)		Function (List II)
A.	Plasma	1.	Transport of oxygen and carbon dioxide
B.	Red Blood Cells (RBCs)	2.	Clot formation
C.	White Blood Cells (WBCs)	3.	Regulation of blood volume and pressure
D.	Platelets	4.	Defense against pathogens

1. A-1, B-2, C-4, D-3
2. A-3, B-1, C-4, D-2
3. A-3, B-1, C-2, D-4
4. A-2, B-4, C-1, D-3

108. Hormones which interact with membrane-bound receptors:

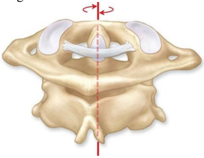
I:	normally do not enter the target cell
II:	generate second messengers which in turn regulate cellular metabolism
III:	include steroid hormones and iodothyronines

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

109. During glycolysis, ATP is directly synthesised in which step?

1.	Conversion of glucose to glucose-6-phosphate
2.	Conversion of fructose-6-phosphate to fructose-1,6-bisphosphate
3.	Conversion of glucose to fructose-6-phosphate
4.	Conversion of BPGA to 3-phosphoglyceric acid (PGA)

110. What type of synovial joint is shown in the given diagram?



1.	Pivot	2.	Gliding
3.	Saddle	4.	Ball and socket

111. All the following are steroid hormones except:
1. cortisol
2. testosterone
3. adrenocorticotrophic hormone
4. estradiol

112. The atrio-ventricular node (AVN) in the human heart is located:

1.	in the right upper corner of the right atrium.
2.	in the lower left corner of the right atrium close to the atrio-ventricular septum.
3.	in the lower right corner of the left atrium close to the atrio-ventricular septum.
4.	in the interventricular septum

113. Consider the given two statements:

Assertion(A)	Monocotyledonous roots do not undergo any secondary growth.
Reason (R):	There are usually more than six (polyarch) xylem bundles in the monocot root.

1.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain the **(A)**.
2.	**(A)** is True but **(R)** is False.
3.	Both **(A)** and **(R)** are True and **(R)** correctly explains the **(A)**.
4.	**(A)** is False but **(R)** is True.

114. Water containing cavities within the vascular bundles is a feature of the anatomy of a:
1. Monocot stem
2. Dicot stem
3. Monocot root
4. Dicot root

115. Match the following major plasma proteins with their functions:

	Column A		Column B
1.	Albumin	A.	Blood clotting
2.	Globulin	B.	Colloid osmotic pressure
3.	Fibrinogen	C.	Immune responses

Options:

1.	1-B, 2-A, 3-C	2.	1-A, 2-B, 3-C
3.	1-C, 2-A, 3-B	4.	1-B, 2-C, 3-A

116. Consider the given two statements:

Assertion(A):	Individuals with type AB blood are universal recipients for blood transfusions.
Reason(R):	They possess neither A nor B antigens on their red blood cells.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False but **(R)** is True.

117. What type of meristem helps grasses regenerate parts removed by the grazing herbivores?

1.	Apical	2.	Intercalary
3.	Secondary	4.	Lateral

118. What type of arthritis is a metabolic disorder in which an abnormal amount of uric acid crystals is deposited in the joint?

1.	gouty	2.	rheumatoid
3.	osteo	4.	psoriatic

119. Select the correct statement with respect to disorders of muscles in humans.

1.	Failure of neuromuscular transmission in myasthenia gravis can prevent normal swallowing
2.	Accumulation of urea and creatine in the joints causes their inflammation
3.	An overdose of vitamin D causes osteoporosis
4.	Rapid contractions of skeletal muscles causes muscle dystrophy

120. Red muscle fibres when compared to the white muscle fibres have a higher:

I:	number of mitochondria
II:	number of blood capillaries
III:	myoglobin content

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

121. Consider the following statements:

I:	70% of ventricular filling is due to the passive opening of AV valves.
II:	Lub is caused by the simultaneous closure of both AV valves.
III:	Dub is caused by simultaneous closure of both the semilunar valves.

Which of the above statements are true?

1.	**I** and **II** only	2.	**I** and **III** only
3.	**II** and **III** only	4.	**I**, **II**, and **III**

122. The thyroid gland:

I:	is composed of two lobes interconnected with a thin flap of connective tissue called isthmus.
II:	is composed of follicles and stromal tissues.
III:	is located in the mediastinum, ventral to aorta.

1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

123. Select the correct statement.
1. Glucagon is associated with hypoglycemia.
2. Insulin acts on pancreatic cells and adipocytes.
3. Insulin is associated with hyperglycemia.
4. Glucocorticoids stimulate gluconeogenesis.

124. Oxidation of one molecule of NADH gives rise to - _____ molecules of ATP, while that of one molecule of $FADH_2$ produces _____ molecules of ATP.

1.	3 and 2 respectively	2.	2 and 3 respectively
3.	1 and 2 respectively	4.	2 and 1 respectively

125. Hormones that promote gluconeogenesis, proteolysis, and lipolysis, and are involved in stress response are secreted by:
1. Pancreas
2. Thyroid gland
3. Adrenal cortex
4. Parathyroid gland

126. Which of the following takes place in the step where succinyl coenzyme A is converted to succinic acid in the TCA cycle?
1. One molecule of ATP is directly synthesized.
2. NAD+ is reduced.
3. FAD is reduced.
4. One molecule of GTP is synthesized.

127. In aerobic respiration, passing on of the electrons removed as part of the hydrogen atoms to molecular O_2 with simultaneous synthesis of ATP, is associated with:
1. Outer mitochondrial membrane
2. Inner mitochondrial membrane
3. Mitochondrial matrix
4. Cytosol

128. Which of the following is a characteristic feature of the epidermal tissue system?

1.	Presence of chloroplasts in all epidermal cells
2.	Formation of vascular bundles in the epidermal layer
3.	Development of lateral roots during secondary growth within the epidermis
4.	Presence of a thick, waxy cuticle covering the epidermis

129. Rh incompatibility can lead to grave consequence in cases where:

1.	an Rh negative mother is carrying an Rh positive foetus
2.	an Rh negative mother is carrying an Rh negative foetus
3.	an Rh positive mother is carrying an Rh positive foetus
4.	an Rh positive mother is carrying an Rh negative foetus

130. Companion cells are closely associated with:

1.	Sieve elements	2.	Vessel elements
3.	Trichomes	4.	Guard cells

131. Consider the given two statements:

Assertion(A):	Synovial joints allow considerable movement.
Reason(R):	Synovial joints are characterised by the presence of a fluid filled synovial cavity between the articulating surfaces of the two bones.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.
3.	Both **(A)** and **(R)** are False.
4.	**(A)** is True and **(R)** is False.

132. Adrenaline and noradrenaline:
I: increase alertness
II: dilate pupil
III: cause piloerection (raising of hairs)
1. Only **I** and **II** are correct
2. Only **I** and **III** are correct
3. Only **II** and **III** are correct
4. **I**, **II** and **III** are correct

133. What are the primary functions of the thyroid gland?

1.	Regulation of calcium and phosphate levels
2.	Regulation of metabolic rate, brain development, and maintenance of calcium levels
3.	Control of pigment synthesis and breakdown
4.	Regulation of blood sugar levels and hunger

134. The correct differential white blood cell count [in order from most common to least common] will be:

1.	basophils, eosinophils, monocytes, lymphocytes, neutrophils
2.	lymphocytes, neutrophils, monocytes, eosinophils, basophils
3.	neutrophils, monocytes, lymphocytes, basophils, eosinophils
4.	neutrophils, lymphocytes, monocytes, eosinophils, basophils

135. What is a function of the pericycle in plants?

1.	Photosynthesis
2.	Storage of food
3.	Secondary growth initiation
4.	Protection from pathogens

136. Consider the given two statements:

Statement A:	The primary meristem in plants appears later than the secondary meristem and is responsible for producing secondary tissues.
Statement B:	Lateral meristems such as fascicular vascular cambium and cork cambium are cylindrical in shape and contribute to the formation of secondary tissues.

Which of the following is correct?

1.	Only **Statement A** is correct.
2.	Only **Statement B** is correct.
3.	Both **Statement A** and **Statement B** are correct.
4.	Both **Statement A** and **Statement B** are incorrect

137. During glycolysis, the step in which 3-phosphoglyceraldehyde (PGAL) is converted to 1,3-bisphosphoglycerate involves:

1.	Utilization of NAD^+ as an oxidizing agent to form $NADH + H^+$.
2.	Conversion of ATP to ADP.
3.	Production of CO_2 as a by-product.
4.	Reduction of FAD^+ to $FADH_2$.

138. Products of the Krebs cycle include:
I. carbon dioxide
II. NADH
III. $FADH_2$
1. Only I and II
2. Only I and III
3. Only II and III
4. I, II and III

139. Which of the following hormones does not act on target cells through a second messenger?
1. Insulin
2. Adrenaline
3. PTH
4. Thyroxine

140. A motor unit is composed of:

1.	a motor neuron and all of the skeletal muscle fibers innervated by the neuron's axon terminals.
2.	all motor neurons and all of the skeletal muscle fibers innervated by them in a muscle.
3.	a motor neuron and one skeletal muscle fiber innervated by the neuron.
4.	a motor neuron and a sensory neuron.

141. Which component of the ECG represents ventricular repolarization?
1. P wave
2. T wave
3. PR interval
4. QRS complex

142. The stage of cellular respiration that directly involves oxygen is:

1.	glycolysis
2.	at the end of ETC
3.	Krebs cycle
4.	the phosphorylation of ADP to form ATP

143. Identify the incorrect statement regarding guard cells in the epidermal tissue system?

1.	They enclose stomata.
2.	They are bean shaped in grasses.
3.	The outer walls are thin and the inner walls are highly thickened.
4.	They possess chloroplasts.

144. When fatty acids are being used as respiratory substrates, they enter cellular respiration as:
1. Glucose
2. Glyceraldehyde Phosphate
3. Pyruvate
4. Acetyl Coenzyme A

145. During oxidation within a cell, all the energy contained in respiratory substrates:
1. is released free into the cell
2. is released in a single step
3. is released in a series of slow stepwise reactions
4. gets converted to usable energy

146. What does the 'P' wave in an ECG represent?
1. Ventricular depolarization
2. Atrial repolarization
3. Ventricular repolarization
4. Atrial depolarization

147. The H-zone in the skeletal muscle fibre is due to:

1.	The central gap between myosin filaments in the A-band.
2.	The central gap between actin filaments extends through myosin filaments in the A-band.
3.	Extension of myosin filaments in the central portion of the A-band
4.	The absence of myofibrils in the central portion of the A-band.

148. The pituitary gland

I:	is divided anatomically into an adenohypophysis and a neurohypophysis.
II:	Adenohypophysis consists of two portions, pars distalis and pars intermedia.

1.	Both **I** and **II** are correct
2.	Both **I** and **II** are incorrect
3.	**I** is correct, **II** is incorrect
4.	**I** is incorrect, **II** is correct

149. Trichomes in stems can perform all the following functions except:

1.	Generate secretory products to cause sting.
2.	Reduce herbivory.
3.	Bring about greater water loss through extended surface area of the epidermis.
4.	Provide shade.

150. For each molecule of glucose, the number of molecules of Acetyl CoA entering the Citric Acid Cycle is:

1.	1	2.	2
3.	3	4.	4

151. Lack of relaxation between successive stimuli in sustained muscle contraction is known as?
1. fatigue
2. tetanus
3. tonus
4. spasm

152. All the following can act as second messengers for hormones in our body except:
1. calcium ions
2. potassium ions
3. Inositol phosphate
4. cyclic adenosine monophosphate

153. Correct comparisons between fermentation and aerobic respiration will include:

I:	Fermentation accounts for only a partial breakdown of glucose, whereas in aerobic respiration, it is completely degraded to CO_2 and H_2O.
II:	In fermentation, there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid, whereas many more molecules of ATP are generated under aerobic conditions.
III:	NADH is oxidised to NAD^+ rather slowly in fermentation, however, the reaction is very vigorous in case of aerobic respiration.

1. Only **I** and **II**
2. Only **I** and **III**
3. Only **II** and **III**
4. **I**, **II** and **III**

154. Consider the given two statements – an Assertion [A] and a Reason [R] – and select the correct option as your answer:

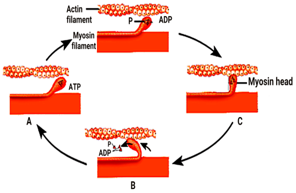
A:	Monocots do not form secondary tissues.
R:	The vascular bundles have no cambium present in them.

1.	Both Assertion and Reason are true but Reason does not correctly explain Assertion
2.	Assertion is true; Reason is false
3.	Both Assertion and Reason are true and Reason correctly explains Assertion
4.	Assertion is false; Reason is true

155. In the human ribcage, true ribs are defined as:

1.	Floating ribs that do not connect to the sternum
2.	Ribs connected indirectly to the sternum via cartilage of other ribs
3.	Ribs directly attached to the sternum via costal cartilage
4.	Ribs that do not articulate with the vertebral column

156. The given figure shows stages in cross bridge formation, rotation of head and breaking of cross bridge during muscle contraction. Select the correct match for A, B and C from the codes given:



I:	Sliding or rotation	A
II:	Breaking of cross bridge	B
III:	Formation of cross bridge	C

Codes:

	I	**II**	**III**
1.	B	A	C
2.	B	C	A
3.	A	B	C
4.	C	A	B

157. Consider the two statements:

Assertion (A):	Any deviation from the normal shape of ECG indicates a possible abnormality or disease.
Reason (R):	The ECGs obtained from different individuals have roughly the same shape for a given lead configuration.

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**.
2.	**(A)** is True but **(R)** is False.
3.	Both **(A)** and **(R)** are True but **(R)** does not correctly explain **(A)**.
4.	**(A)** is False but **(R)** is True.

158. In a dicotyledonous root, the innermost layer of the cortex, which is composed of barrel-shaped cells without intercellular spaces, is called:
1. Epiblema
2. Endodermis
3. Pericycle
4. Pith

159. What is the function of thrombokinase in blood clotting?

1.	Thrombokinase directly converts fibrinogen into fibrin, which forms a clot.
2.	Thrombokinase is an enzyme complex that converts prothrombin into thrombin, which subsequently converts fibrinogen into fibrin.
3.	Thrombokinase prevents clot formation by inhibiting thrombin activity.
4.	Thrombokinase dissolves clots by breaking down fibrin into smaller fragments.

160. Which of the following statements about cork cambium is incorrect?

1.	It forms a secondary cortex on its outside
2.	It forms a part of periderm
3.	It is responsible for the formation of lenticels
4.	It is a couple of layers thick

161. Thyroid gland, in humans, produces:
I: Iodothyronines
II: Peptide hormone
1. Only **I** is correct
2. Only **II** is correct
3. Both **I** and **II** are correct
4. Both **I** and **II** are incorrect

162. Consider the given two statements:

Statement I:	Acetyl-CoA, is the starting point for the citric acid cycle.
Statement II:	Acetyl-CoA may also be obtained from the oxidation of fatty acids.

1. Statement I is correct; Statement II is correct
2. Statement I is incorrect; Statement II is correct
3. Statement I is correct; Statement II is incorrect
4. Statement I is incorrect; Statement II is incorrect

163. Consider the two statements:

Statement I:	Each rib is a thin flat bone connected ventrally to the vertebral column and dorsally to the sternum.
Statement II:	Each rib has two articulation surfaces on its ventral end and is hence called bicephalic.

1. **Statement I** is correct; **Statement II** is correct
2. **Statement I** is correct; **Statement II** is incorrect
3. **Statement I** is incorrect; **Statement II** is correct
4. **Statement I** is incorrect; **Statement II** is incorrect

164. Which of the following has an ATPase activity?
1. Myosin head
2. Myosin tail
3. Troponin T
4. Tropomyosin

165. When the heart muscle is suddenly damaged by an inadequate blood supply, the condition is called as

1.	Heart Attack	2.	Heart Failure
3.	Cardiac Arrest	4.	Angina pectoris

166.

Assertion(A):	The first seven pairs of ribs in humans are called as true ribs.
Reason(R):	The first seven pairs of ribs in humans directly articulate with the sternum through their costal cartilages.

1.	Both **(A)** and **(R)** are True and **(R)** is the correct explanation of **(A)**.
2.	Both **(A)** and **(R)** are True but **(R)** is not the correct explanation of **(A)**.
3.	**(A)** is True but **(R)** is False.
4.	**(A)** is False and **(R)** is True.

167. A cardiovascular centre that can moderate the cardiac function through the autonomic neural system is located in the:

1.	Pons varoli	2.	Medulla oblongata
3.	Hypothalamus	4.	Cerebrum

168. Which meristem is responsible for the regeneration of parts of grasses removed by the grazing herbivores?
1. Apical meristem
2. Intercalary meristem
3. Interfascicular cambium
4. Fascicular vascular cambium

169. Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body:

(a) (b) (c)

Name of muscle/location

1.	(a) Skeletal - Triceps (b) Smooth - Stomach (c) Cardiac - Heart	2.	(a) Skeletal - Biceps (b) Involuntary - Intestine (c) Smooth - Heart
3.	(a) Involuntary - Nose tip (b) Skeletal - Bone (c) Cardiac - Heart	4.	(a) Smooth - Toes (b) Skeletal - Legs (c) Cardiac - Heart

170. What is a key difference between red muscle fibers and white muscle fibers?

1.	Red muscle fibers are adapted for endurance and have a high myoglobin content, while white muscle fibers are designed for short bursts of activity and contain less myoglobin.
2.	White muscle fibers are rich in myoglobin, whereas red muscle fibers are not.
3.	Red muscle fibers are smaller in diameter than white muscle fibers.
4.	White muscle fibers have more mitochondria than red muscle fibers.

171. During cardiac cycle:

1.	all the four chambers of heart can never be in a relaxed state simultaneously.
2.	as the tricuspid valve gets open, blood from the pulmonary artery flows into the right ventricle.
3.	all the four chambers of heart can never be in a contracted state simultaneously.
4.	as the bicuspid valve gets open, blood from the aorta flows into the left ventricle.

172. In anaerobic glycolysis, how many net ATP molecules are generated per molecule of glucose in plant cells?
1. 2 ATP
2. 4 ATP
3. 6 ATP
4. 8 ATP

173. The secretion of glucocorticoids from the adrenal cortex is primarily regulated by:

1.	The blood glucose and sodium levels.
2.	The direct stimulation of adrenal glands by the autonomic nervous system.
3.	The release of adrenocorticotropic hormone (ACTH) by the anterior pituitary.
4.	The negative feedback inhibition of epinephrine release.

174. Consider the given two statements:

Assertion (A):	Presence of oxygen is vital for aerobic cellular respiration
Reason (R):	Oxygen is required in all steps from glycolysis to electron transport involved in aerobic cellular respiration

1.	Both **(A)** and **(R)** are True and **(R)** correctly explains **(A)**
2.	Both **(A)** and **(R)** are True and **(R)** does not correctly explain **(A)**
3.	**(A)** is True; **(R)** is False
4.	**(A)** is False; **(R)** is True

175. Calcium is important in skeletal muscle contraction because it:

1.	binds to troponin to remove the masking of active sites on actin for myosin.
2.	activates the myosin ATPase by binding to it.
3.	detaches the myosin head from the actin filament.
4.	prevents the formation of bonds between the myosin cross bridges and the actin filament.

176. Match List-I with List-II:

	List-I		List-II
A.	Tricuspid valves	I.	Guards the opening between right atrium and right ventricle
B.	Sino-atrial node	II.	Guards the opening between the left atrium and the left ventricle
C.	Mitral valves	III.	Guards the opening of right and left ventricles into pulmonary artery and aorta, respectively
D.	Semilunar valves	IV.	Cardiac musculature in the right upper corner of right atrium

Choose the correct answer from the options given below:
1. A-III, B-IV, C-II, D-I
2. A-II, B-I, C-IV, D-III
3. A-III, B-IV, C-I, D-II
4. A-I, B-IV, C-II, D-III

177. Glucocorticoids are involved in all of the following, except
1. Producing anti-inflammatory reactions
2. Immunosuppressive function
3. Inhibiting gluconeogenesis, lipolysis, and proteolysis
4. Stimulating RBC production

178. Inflammation of joints due to accumulation of uric acid crystals is called as:
1. Rheumatoid arthritis
2. Gouty arthritis
3. Osteoarthritis
4. Psoriatic arthritis

179. A plant with a fibrous root system and leaves with parallel venation would lack
I: A vascular cambium.
II: Secondary xylem and phloem.
III: A cork cambium.

1.	**I**, **II**, and **III**	2.	**I** and **II**
3.	**II** and **III**	4.	**I** and **III**

180. The anatomy of a dicot root typically includes:

1.	A single layer of epidermis with root hairs
2.	A large central pith
3.	Scattered vascular bundles
4.	Secondary growth absent